Bioinformatics and Systems Biology (joint degree)

Vrije Universiteit Amsterdam - Faculty of Science - Bioinformatics and Systems Biology (joint degree) - 2025-2026

Vast amounts of data have been collected through genomics initiatives. They provide a golden opportunity to research the secrets of life, to understand more of its complexities, to improve quality of life and to conquer major diseases. Converting this huge volume of data into real understanding is the basic challenge of Bioinformatics and Systems Biology research. Which computational techniques are available to make predictions about health outcomes and decide on a personalised treatment for a patient? How and when can we use such techniques? How can we set up experiments and models in such a way that we start to understand the complex underlying systems?

The first year in particular devotes a great deal of attention to the fundamentals of both bioinformatics and systems biology. Students with a specialisation in bioinformatics will extend their knowledge in sequence analysis, full genome sequencing, protein structure prediction and simulation, and medical applications of bioinformatics. Students with a specialisation in systems biology will extend their knowledge in models of biological networks, experimental verifications of models, relevance of systems biology to research and synthetic biology.

In the second and final year students perform one or two research projects.

The programme is shared between VU Amsterdam and the University of Amsterdam (UvA), combining the resources of two excellent universities.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Joint degree
Teaching and examination regulation 2025-2026
More information about this programme

Bioinformatics and Systems Biology JD compulsory courses 4
Bioinformatics and Systems Biology JD specialisation Systems Biology 4
Bioinformatics and Systems Biology JD specialisation Systems Biology compulsory courses 4
Bioinformatics and Systems Biology JD specialisation Bioinformatics 4
Bioinformatics and Systems Biology JD specialisation Bioinformatics compulsory courses 5
Bioinformatics and Systems Biology JD constrained choice 5
Bioinformatics and Systems Biology JD optional courses 5
Courses 6
Algorithms in Sequence Analysis 6
Basic Models of Biological Networks 7
Bioinformatics for Translational Medicine 9
Biomolecular Simulations 11
Biosystems Data Analysis 11
Brain Imaging 13
Computational Biology 13
Data Mining Techniques 14
Deep Learning 15
Drug Target Biochemistry and Signaling 17
Evolutionary Computing 18
Fundamentals of Bioinformatics 19
Image Processing and Quantitative Data Analysis 21
Introduction to Systems Biology 23
Knowledge Representation 24
Machine Learning for the Quantified Self 25
Major Bioinformatics Research Project BSB 27
Major Systems Biology Research Project BSB 28
Minor Research Project M BSB 30
Protein Science 31
Quantitative Single-cell Biology 33
Scientific Machine Learning 34
Statistics with R 36
Structural Bioinformatics 37
Systems Biology in Practice 39

Bioinformatics and Systems Biology JD compulsory courses

Description

The compulsory courses in the master Bioinformatics and Systems Biology prepare you for the deeper dive into these topics, by laying the groundwork in molecular biology, programming in python and R, calculus, linear algebra, machine learning, statistics and multivariate data analysis.

It is recommended to complete all compulsory courses before you can start a minor or major research project. For the minor project you need to complete at least three and for the major project at least five out of compulsory or specialization courses, before you are allowed to start your project.

See the TER for the specific rules.

Courses

Course Name	Period	Credits	Code
Fundamentals of Bioinformatics	P1	6EC	X_405052
Introduction to Systems Biology	P1	6EC	X_428565
Statistics with R	P2	6EC	X_418156
Biosystems Data Analysis	P3	6EC	XM_0078

Bioinformatics and Systems Biology JD specialisation Systems Biology

Description

Within the specialisation Systems Biology you cover topics such as metabolic networks, reaction kinetics, generation and analysis of molecular high-throughput data, and advanced mathematical modelling. It is possible to follow the Systems Biology and Bioinformatics specialisation courses at the same time if you want to complete both specialisations.

Programme Sections

- Bioinformatics and Systems Biology JD specialisation Systems Biology compulsory courses

Bioinformatics and Systems Biology JD specialisation Systems Biology compulsory courses

Description

You must complete all compulsory courses within at least one of both specialisations.

It is recommended to complete all compulsory courses taught within the Systems Biology specialisation before you can start your major Systems Biology research project. For the minor project you need to complete at least three and for the major project at least five out of compulsory or specialization courses, before you are allowed to start your project.

See the TER for the specific rules.

Courses

Course Name	Period	Credits	Code
Major Systems Biology Research Project BSB	Ac. Year (September)	30EC	XM_0104
Basic Models of Biological Networks	P4	6EC	X_418154
Systems Biology in Practice	P5	6EC	XM_0079

Bioinformatics and Systems Biology JD specialisation

Bioinformatics

Description

Within the specialisation Bioinformatics you cover topics such as structural bioinformatics, advanced sequence analysis algorithms, machine learning and validation and analysis of molecular high-throughput data. It is possible to follow the Bioinformatics and Systems Biology specialisation courses at the same time if you want to complete both specialisations.

Programme Sections

- Bioinformatics and Systems Biology JD specialisation Bioinformatics compulsory courses

Bioinformatics and Systems Biology JD specialisation Bioinformatics compulsory courses

Description

You must complete all compulsory courses within at least one of both specialisations.

It is recommended to complete all compulsory courses taught within the Bioinformatics specialisation before you can start your major Bioinformatics research project. For the minor project you need to complete at least three and for the major project at least five out of compulsory or specialization courses, before you are allowed to start your project.

See the TER for the specific rules.

Courses

Course Name	Period	Credits	Code
Major Bioinformatics Research Project BSB	Ac. Year (September)	30EC	XM_0103
Algorithms in Sequence Analysis	P2	6EC	X_405050
Structural Bioinformatics	P4	6EC	X_405019
Bioinformatics for Translational Medicine	P5	6EC	X_405092

Bioinformatics and Systems Biology JD constrained choice

Description

You must take one of the two constrained choice courses; you are free to choose which one.

Courses

Course Name	Period	Credits	Code
Image Processing and Quantitative Data Analysis	P6	6EC	XM_0085
Quantitative Single-cell Biology	P6	6EC	XM_0084

Bioinformatics and Systems Biology JD optional courses

Description

Select sufficient courses from the below options to fill your elective space, with a minimum of 12 EC (depending on your specialisation) and a maximum of 36 EC.

If you want to take an optional course that is not on the list, you must request permission from the Examination Board beforehand.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Minor Research Project M BSB	Ac. Year (September)	18EC	XM_0072
Evolutionary Computing	P1	6EC	X_400111
Protein Science	P1	6EC	AM_470145
Algorithms in Sequence Analysis	P2	6EC	X_405050
Deep Learning	P2	6EC	XM_0083
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Knowledge Representation	P2	6EC	XM_0059
Basic Models of Biological Networks	P4	6EC	X_418154
Brain Imaging	P4	6EC	P_MBRIMAG_AI
Scientific Machine Learning	P4	6EC	XM_0138
Structural Bioinformatics	P4	6EC	X_405019
Bioinformatics for Translational Medicine	P5	6EC	X_405092
Biomolecular Simulations	P5	6EC	XMU_0036
Computational Biology	P5	6EC	XMU_0032
Data Mining Techniques	P5	6EC	X_400108
Systems Biology in Practice	P5	6EC	XM_0079
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Courses

Algorithms in Sequence Analysis

Course Code	X_405050
Credits	6
Period	P2
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. S. Pissis
Teaching Staff	dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. S. Pissis, prof. dr. J. Heringa, dr. Z. Işik MSc, H. Verbeek MSc
Teaching method(s)	Computer lab, Lecture

Course Objective

Have you ever wondered how we can track a gene across 3 billion years of evolution? Or how you can use the genome information of a given cancer patient to find out what may be wrong? Sequence alignment can be used to compare genomes, genes or proteins from bacteria all the way to humans, while further derived algorithms may be employed to make a phylogeny (to find out about evolutionary relationships), find a functional motif in a protein sequence, or a viral sequence in a genome. In this course we focus on the most important algorithms for biological sequence analysis that can be applied to real scientific problems in biology.

After completing the course successfully,

- You can demonstrate in-depth knowledge about the theory of sequence analysis methods.
- You know how to implement several of the most important sequence alignment and analysis algorithms.
- You understand the mathematical formulation behind sequence analysis algorithms.
- You will have understanding and skills to apply sequence analysis algorithms to protein and DNA sequences.
- You will have hands-on experience in tackling biological problems using sequence analysis algorithms, including the statistical framework of HMMs and algorithms used in genome sequencing and analysis.
- You can decide which algorithm is best suited for a particular biological sequence analysis problem.
- You can read through scientific and technical literature and learn to translate algorithms described in text and formal mathematical notation into computer code.

Course Content

Theory:

- Dynamic programming
- database searching
- pairwise and multiple alignment
- probabilistic methods including HMMs
- pattern matching
- entropy measures
- evolutionary models
- and phylogeny

Practical:

- Programming (in Python) an alignment algorithm based on dynamic programming;
- Aligning sequencing data from tumors to the human genome and analysing structural variants;
- Programming (in Python) an implementation of HMMs and using it to predict protein domain structure.

Additional Information Teaching Methods

13 lectures: 2 two-hour lectures per week.
13 computer practicals and associated assignments: 2 two-hour hands-on sessions per week.

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment. (assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

The theoretical assessment will be an oral and/or written exam (depending on number of students).

Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Bachelor in any science discipline (including medicine).
Basic programming skills (Python) and an interest in biological problems.

Literature

Course material on Canvas.

Books: Durbin, R., Eddy, S.R., Krogh, A., Mitchison, G.. Biological Sequence Analysis. Cambridge University Press, 1998, 350 pp., ISBN 0521629713.

Recommended reading: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics. Garland Science 2008. ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)

We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. Ideally, this laptop supports a command line shell, Python 3 and a text editor with syntax highlighting -- either standalone (e.g. Atom or Sublime Text) or as part of a simple IDE (e.g. Spyder). As such, we recommend the Anaconda python distribution regardless of operating system, along with PuTTy or PowerShell for Windows users specifically.

The course is taught in English.

Basic Models of Biological Networks

Course Code	X_418154
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.J. Bruggeman
Examiner	prof. dr. F.J. Bruggeman
Teaching Staff	prof. dr. F.J. Bruggeman, dr. G. Stolf Jeuken
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

This course focuses on teaching the creation and analysis of mathematical models for biological systems across various levels of organization, from molecular circuits to cellular systems and potentially ecosystems. The primary modelling formalism explored is ordinary differential equations (ODEs), providing a foundation for understanding biological dynamics. The course aims to equip students with the ability to reproduce models from scientific literature and apply them to solve practical problems. It combines lectures with workgroups involving problem-solving using both computational tools (Python) and pen and paper.

Learning Goals:

- Learn to develop and analyse mathematical models (primarily ODE-based) for diverse biological networks (molecular, cellular, etc.).
- Understand how to represent biological processes, particularly molecular mechanisms like enzyme kinetics and protein regulation, using mathematical formalisms like state-transition diagrams and mass-action kinetics.
- Gain proficiency in simplifying complex mechanisms into net process models suitable for integration into larger system models.
- Develop the skills to reproduce existing models from scientific papers and apply modelling techniques to solve real problems.
- Acquire the intuition necessary to construct useful and informative models by analysing various examples.
- Understand the difference between mechanistic and phenomenological/empirical models and their applications.

Course Content

The course slowly builds on the principles required to make useful models of biological dynamics, starting from first principles:

(1) Fundamentals of Dynamical Systems in Biology:

- Introduction to change, dynamics, systems organization, and the role of energy in biological systems.

(2) Modelling Molecular Circuits:

- Using state-transition diagrams to visualize molecular states and transitions.
- Applying mass-action kinetics to describe the rates of spontaneous molecular reactions and processes within state-transition diagrams.
- Developing mathematical models based on state-transition diagrams.

(3) Net Models and Simplification:

- Modelling enzyme kinetics, including catalytic cycles, rapid-equilibrium approximations, steady-state methods, and cooperative effects.
- Developing equilibrium-binding equations for regulatory proteins (e.g., transcription factors, receptors).
- Reducing complex state-transition diagrams to simpler net process models for use in larger systems.

(4) Models of Larger Systems:

- Applying simplified models to study metabolism, signalling, and signal transduction.
- Introduction to phenomenological and empirical models for systems where detailed mechanisms are unknown or too complex (e.g., cell growth, ecosystems).

(5) Practical Application:

- Solving exercises using Python and pen and paper to reinforce concepts and build modelling skills.

Additional Information Teaching Methods

- Lectures
- Workgroups
- Self-study
- Computer practicals
- Computer modelling tutorials

Method of Assessment

- 80% Written exam
- 20% Computer exam (Python)

You must have a grade higher than 5.5/10 on both exams to pass.

Literature

A course syllabus written by the lecturer

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Mathematics

Bioinformatics for Translational Medicine

Course Code	X_405092
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. E. Bosdriesz
Teaching Staff	dr. E. Bosdriesz, dr. J.M. Urtasun Elizari
Teaching method(s)	Practical, Lecture

Course Objective

Observations from biological high-throughput experiments allow us to improve diagnosis and give a personalized treatment plan for patients. However, integrating data from several sources and using this data for predictions is non-trivial.

This is a theoretical and practical Bioinformatics course on computational methods for Translational Medicine; we will focus on Bioinformatics methods that are used to predict the clinical outcome for patients and analysis methods to obtain deeper understanding of complex diseases, by combining data from various high-throughput experiments such as proteomics, microarrays and next-generation sequencing as well as existing biological databases.

After completing the course successfully,

- You can interpret research results in the context of biology
- You will be aware of Bioinformatics methods that are applicable to the area of Translational Medicine
- You can combine these methods into a creative solution using large-scale biological experiments
- You will have hands-on experience in handling large biological datasets, and will understand the complexity of the biological data both from high-throughput experiments and existing biological databases
- You will be able to work as a team to execute a large, complex biomedical bioinformatics project, and report on the results
- You will be familiar with a few in-depth research topics that lie within the expertise area of several (Bioinformatics) researchers, among others from the VU, UvA, AUMC and NKI

Course Content

Theory

- Computational analysis of molecular profiling techniques, such as WGS, WES, proteomics, RNA sequencing,

and arrayCGH.
- Computational methods applied to these data types, such as: machine learning, normalisation, regularization, feature selection, classification, read mapping, clustering.
- All data analysis is relevant in a clinical setting, for diagnosis, treatment decisions or biomarker discovery.

Practical

- A large assignment for which you have to build a classifier based on molecular profiling data (the precise data-set is to be determined). You need to hand in predictions, write a paper and give a presentation. Note that this is a group project, and to participate in this practical you need to have obtained at least 12 ECT from Master-level courses.
- A smaller assignment that involves analysing and understanding molecular profiling data of another modality than the large classifier assignment.

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- 12 computer practicals (2 two-hour sessions per week)

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment.

Practical assessment (50%):

- Classifier assignment (45%)
- Other data modality assignment (5%)

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Theoretical assessment (50%):

- Oral or written exam (depending on number of course students)
- The exam is based on a selection of 8-10 scientific papers in the field of Bioinformatics & Translational Medicine.

Entry Requirements

Importantly: to participate in the group assignment, you need to have obtained a minimum of 12 ECT of Master level courses. In addition, some basic programming skills (either in R or Python) are required, as well as some basic knowledge on molecular biology. If you are not following the Master Bioinformatics and Systems Biology, it is advisable to first follow the Master course "Fundamentals of Bioinformatics".

Literature

- Course material on Canvas, including slides and recordings of lectures.
- 8-10 scientific papers are provided, and make up the course syllabus.

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

The course is taught in English.

- Compulsory course for students in Master Bioinformatics and Systems Biology
- Optional course for students with a Bachelor in Physics, Chemistry, Mathematics, Computer Science, Biology, or Biomedical Sciences (see requirements below).

Recommended background knowledge

If you are not following the Master Bioinformatics and Systems Biology, it is advisable to first follow the Master course "Fundamentals of Bioinformatics".

Biomolecular Simulations

Course Code	XMU_0036
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. M.L. van Daalen
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/130530

Additional Information Teaching Methods

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Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Biosystems Data Analysis

Course Code	XM_0078
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.A. Westerhuis
Examiner	dr. J.A. Westerhuis
Teaching Staff	dr. D. Molenaar, dr. J.A. Westerhuis, dr. G. Stolf Jeuken
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

The amount of omics data being gathered, analysed, integrated and interpreted increases every year. The number of algorithms, packages and webservices also increases very fast. With the high number of omics features e.g. genes, proteins or metabolites measured, the chances of drawing incorrect conclusions are rather high.

In this course we will discuss properties of the measured data, assumptions of the data analysis methods being used, data preprocessing and pretreatment and validation of the conclusions and interpretations.

Upon completion of this course, students will have reached the following goals

1. Students are able to use R for multivariate analysis of omics data. [KBB 4]
2. Students understand linear algebra concepts that are used in data analysis methods. [KBB 5]

3. Students are able to apply data preprocessing methods (variance stabilization, scaling, transformation), explorative data analysis methods (PCA, Clustering) and supervised data analysis (PCDA, ASCA, Random Forest). [KAR 1]
4. Students are able to evaluate which preprocessing methods, and which data analysis method to use for a given problem. [KAR 5]
5. Students are able to analyse omics data and interpret the results. [KAR 1]
6. Students are able to evaluate the quality of the data analysis models using cross validation and permutation methods. [KAR 4]
7. Students understand how the data analysis methods work theoretically. [KBB5]
8. Students are able to critically review data analysis applications in which the methods taught methods have been used. [KAR 4]

Course Content

In the analysis of biochemical systems, the tendency is to measure more and more of just a few samples, leading to complex multivariate data sets. Multivariate data analysis methods are often used to explore such sets.

This course covers a broad range of multivariate data analysis methods, for data preprocessing (e.g. variance stabilization), for exploration analysis (PCA, Clustering, nonlinear mapping), supervised analysis (ASCA, Classification, Random Forest). Furthermore, validation techniques such as cross validation and permutation testing are introduced and the interpretation of selected features in terms of function and networks is discussed. The course starts with an introduction on the properties of the different types of functional genomics data.

The main goal of this course is to teach students the theoretical properties of the data analysis methods and how to interpret the results of the multivariate methods and how this relates to the biological problem that is studied.

Additional Information Teaching Methods

- 10 lectures (2h each)
- 10 laptop seminars (4h each)
- Self-study for each of the 10 topics

Method of Assessment

There is a data analysis project in which students apply several methods to a dataset after which they have to interpret the results with respect to the statistical properties and the biological properties of the results. There is a theoretical test in which open questions are asked on the data analysis methods, their properties and especially on how to interpret their results. For the final grade, the weights are 1/3 and 2/3 for the project and the theoretical test, respectively. Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Bachelor in any science discipline, with an interest in biological data analysis. Basic level of linear algebra required. When in doubt please contact the coordinator.

Literature

Literature:

- Will be provided through Canvas.

Software:

- R.

Additional Information Target Audience

Admission to the course will depend on capacity, total number of applications, date of registration and background of the individual student. If the number of applications exceeds the capacity of the course, students may have to be selected and priority will be given in the following order:

- First-year students of the master Bioinformatics and Systems Biology (JD UvA and VU).
- Second-year students of the master Bioinformatics and Systems Biology (JD UvA and VU).
- Students of the master Computational Science (JD UvA and VU).
- Students of other master programmes.

Recommended background knowledge

Linear Algebra, Statistics with R

Brain Imaging

Course Code	P_MBRIMAG_AI
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. T.H.J. Knapen
Examiner	dr. T.H.J. Knapen
Teaching Staff	dr. T.H.J. Knapen, mr. S. Zhang MSc, mr. R. van de Klundert MSc
Teaching method(s)	Seminar, Lecture

Course Objective

fMRI is the main method we have of imaging activations of the living human brain, and thereby probe the computational mechanisms underpinning our cognitive repertoire. The goal of this course is to give you an idea of the awesome possibilities and annoying pitfalls of functional brain imaging, but most importantly, to give you a solid foundation for further learning. At the end of the course, you will be able to devise a valid fMRI experiment, and perform the basic analysis on the resulting data using state-of-the-art open science and open source tools.

Course Content

In this course we will teach you the ins and outs of brain imaging, that is, fMRI. We will teach you everything from the basics of signal analysis, to experimental design, to statistics. Some of the newest cutting-edge techniques, including pattern classification analysis, connectivity modelling, and resting state network analysis, are also discussed.

Additional Information Teaching Methods

Every week, there will be one or two lectures, interspersed with 3 to 4 practicals/'werkcolleges' in total. The course is broadly divided into two parts; the first half of the course serves to teach you the very basics of signal analysis and experimentation. We believe this basis is necessary to later start to think independently and academically about research in your future field. In this first phase of the course the weekly lecture will treat theory while the practicals will allow you to wet your toes with this material. This way we try to combine theory and practice.

In the second half of the course, you will already know a lot about what Brain Imaging entails. Then, we will switch gears a bit, and teach you what's going on in the neuroimaging field right now. That means that during the weekly lecture we will use research articles to illustrate the state of the art. In the practicals we'll move towards letting you perform an entire fMRI analysis yourselves. In this second part of the course we'll also focus more and more on recent articles that show us the state-of-the-art in neuroimaging.

Method of Assessment

Final Exam, open-end & MC questions 70%
Quizzes on Canvas 20%
Perusall reading participation 10%

Entry Requirements

Fluency in the Python programming language.

Literature

We will use articles describing current research.

Additional Information

Prior knowledge of Python programming and statistics is important for following the practicals.

Computational Biology

Course Code	XMU_0032
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. M.L. van Daalen
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-course/course/130582

Additional Information Teaching Methods

-

Method of Assessment

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Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Data Mining Techniques

Course Code	X_400108
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M. Hoogendoorn
Examiner	prof. dr. M. Hoogendoorn
Teaching Staff	dr. S. Yu, prof. dr. M. Hoogendoorn, B. Borsboom
Teaching method(s)	Computer lab, Lecture

Course Objective

The aim of the course is that students acquire data mining knowledge and skills that they can apply in a business environment. More precisely, the following learning goals are distinguished:

1. Understand the data mining process
2. Understand exploratory data analysis
3. Understand feature engineering
4. Understand classification techniques
5. Understand regression techniques
6. Understand association rules and recommender systems
7. Understand big data engineering and big data infrastructures
8. Understand current state of the art in data mining
9. Understand ethical concerns and evaluation of results
10. Learning to argue for choices (rationale)

The course if focused on the Dublin descriptors Applying Knowledge and Understanding (since the aim is to improve the practical skills); Making judgements (which technique is appropriate, how to best apply Data Mining for a specific case); Communication skills (how to report on your approach, choices and your results), and Learning skills (able to find new relevant techniques, assess their suitability, etc.).

How the aims are to be achieved: Students will acquire knowledge and skills mainly through the following: an overview of the most common data mining algorithms and techniques (in lectures), a survey of typical and interesting data mining applications, and practical assignments to gain "hands on" experience. The application of skills in a business environment will be simulated through various assignments of the course.

Course Content

The course is intended to introduce Data Mining Techniques to students that are new to the field as well as to more experienced students. The main aim is to gain a more practical perspective towards Data Mining Techniques/Machine Learning. Lectures will cover more basic things for those new to the field (general introduction into Data Mining, classical algorithms such as decision trees, association rules, neural networks, ensemble learning, etc.) and on top will discuss advanced topics including deep learning, recommender systems, big data infrastructures, and text mining. A number of successful applications in the area will also be discussed. In addition to lectures, there will be an extensive practical part, where students will experiment with various data mining algorithms and data sets. The grade for the course will be based on these practical assignments.

Additional Information Teaching Methods

Lectures (h) and practical sessions (pra). Lectures are planned to be interactive: there will be small questions, etc.

Method of Assessment

Practical assignments. There will be two assignments done in groups of three. For the first assignment there is a choice: going for a basic version of the assignment (suited for those new to the domain) or a more advanced one (for students with more experience). The second assignment is the same for all and will involve an in-class Kaggle competition. The first assignment counts for 40% and the second for 60%.

An interview will take place on an individual student level to make sure that (1) each student understands the content of the graded material, and (2) has made a sufficient contribution to the group work. Only when this check is passed will a final grade be given. Furthermore, the grade of both assignments needs to be sufficient to pass the course. Finally, attendance of the final lecture (in which the final ranking of the Kaggle competition is announced and selected groups of students will be asked to present their solution) is mandatory as it contributes to the learning objectives of the course.

There is an option to resit one of the assignments when the assignment has been submitted but an insufficient grade is obtained (only when the other assignment grade is sufficient). This involves improving your assignment to at least a sufficient level within two weeks after having received your grade for the second assignment. The maximum grade of the resit is a 6.0.

Literature

Optional: Ian H. Witten, Eibe Frank, Mark A. Hall, and Christopher J. Pal. Data Mining: Practical Machine Learning Tools and Techniques (Fourth Edition). Morgan Kaufmann, 2016 978-0128042915.

Additional Information Target Audience

Master Artificial Intelligence, Master Computer Science, Master Bioinformatics and Systems Biology, Master Business Analytics

Recommended background knowledge

1. Statistics
2. Programming Skills

Deep Learning

Course Code	XM_0083
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. S. Yu
Examiner	dr. S. Yu
Teaching Staff	dr. S. Yu, dr. M. Cochez, dr. P. Bloem
Teaching method(s)	Practical, Lecture

Course Objective

1. Knowledge and understanding

- Be able to explain the main components of a deep learning architecture: fully-connected layers, convolutional layers, activation functions, LSTM, and GRU.
- Be able to explain and derive the backpropagation algorithm.
- Be able to explain different commonly used deep architectures.
- Be able to explain Variational Autoencoders and unsupervised representation learning.
- Be able to explain the difference between discriminative and generative models.

2. Applying knowledge and understanding

- How before-mentioned layers work and where to use them.
- How to formulate a neural network for a specific problem.
- How to analyze the performance of a neural network.
- What neural network fits best for a given problem.

3. Making judgments

- What deep learning model to use for a given problem (e.g., generative vs discriminative).
- What layers of a neural network are suitable for a given problem.

4. Communication skills

- Presenting an analysis in a written form (a short report) for each assignment.

5. Learning skills

- Able to read (some) state-of-the-art papers.
- Able to develop (to some extent) available deep learning libraries.

Course Content

Deep learning becomes the leading learning and modeling paradigm in machine learning. During this course, we will present basic components of deep learning, such as:

- different layers (e.g., linear layers, convolutional layers, pooling layers, recurrent layers);
- non-linear activation functions (e.g., sigmoid, ReLU);
- backpropagation;
- learning algorithms (e.g., ADAM);
- other (e.g., dropout).

Further, we will show how to build deep architectures like AlexNet and ResNet. We will explain potential pitfalls and possible solutions, e.g., by using residual connections and dense architectures.

After discussing discriminative models, we will turn them into generative models. We will start with linear latent variable models like the probabilistic PCA. Then we will discuss a non-linear version of the pPCA, namely, Variational Auto-Encoders (VAEs). Both pPCA and VAE are so-called prescribed models that require formulating the likelihood function. Next, we will introduce unsupervised representation learning, different ways to improve VAEs, and how to obtain disentangled representations.

At the end of the course, we will outline recent developments in deep learning. Namely, we will discuss Reinforcement Learning and Deep Reinforcement Learning. In the end, we will introduce generalization and explainabilility - two fundamental topics of deep learning.

Additional Information Teaching Methods

The course consists of two parts: a digital exam and practical assignments. The digital exam is supported by lectures (two/three per week). The assignments are supported by practical sessions led by TAs.

The first two assignments are carried out individually and the remaining assignments are done in small groups. No resit is possible for the practical assignments.

Method of Assessment

Final exam (50%) and practical assignments (50%). There are 4 assignments (2 of them are done individually; 2 of them are done in groups).

The final exam must be passed with a sufficient grade (equivalent to a grade of 5.5 or higher).

There is a resit for the exam. The practical assignments cannot be redone.

Literature

The literature will be made available on Canvas.
Two suggested books:

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). "Deep learning". MIT press.
- Tomczak, J. M. (2022). "Deep Generative Modeling". Springer, Cham.

Additional Information Target Audience

Master Artificial Intelligence
Master Business Analytics

Recommended background knowledge

- Calculus
- Linear Algebra
- Statistics & Probability Theory
- Programming (Python)
- Machine Learning

Drug Target Biochemistry and Signaling

Course Code	XM_0111
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M.J. Smit
Examiner	prof. dr. M.J. Smit
Teaching Staff	prof. dr. M.J. Smit, dr. M.H. Siderius
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Deepening of the knowledge of the biochemical aspects of drug targets and their role in cellular signaling in health and disease, with the aim to understand their importance in modern pharmacology.

Course Content

This course is designed for students with a background in the field of pharmaceutical and life sciences (or have followed Principles of Pharmaceutical Sciences) and an interest in the biological and pharmacological aspects of drug discovery. The course will focus on biochemical properties, including structures, of the main families of drug targets and their involvement in cellular signaling networks in health and disease (e.g. CNS-related disorders, cancer). Different strategies to identify, validate and modulate drug targets (e.g. ‘omics’ approaches, siRNA, CRISPR/Cas9, biologicals) will be addressed. New concepts in the architecture of signaling networks and mechanisms of signal transfer will be discussed.

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Written examination and assignments

Literature

Primary literature and handouts.
'Cell signaling', second edition. Authors: Wendell Lim and Bruce Mayer ISBN: 978-0-367-27937-0

Additional Information Target Audience

1DDS, 1BMS, MSc Bioinformatics and Systems Biology with Life Science background/BSc

Recommended background knowledge

BSc Pharmaceutical Sciences or equivalent or Principles of Pharmaceutical Sciences or BSc in Life Sciences

Evolutionary Computing

Course Code	X_400111
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. A.E. Eiben
Examiner	prof. dr. A.E. Eiben
Teaching Staff	prof. dr. A.E. Eiben, mr. J.M. di Matteo, dr. K. da Silva Miras de Araujo
Teaching method(s)	Seminar, Lecture

Course Objective

This course has a threefold objective:

1. (Knowledge & Understanding) To learn about computational methods based on Darwinian principles of evolution.
2. (Applying knowledge and understanding, Communication) To use such methods as problem solvers and as simulation tools; to determine adequate algorithm setups depending on the problem at hand.
3. (Applying knowledge and understanding, Lifelong learning skills) To gain hands-on experience in performing computational experiments with evolutionary algorithms.

Course Content

This course is about constructing, applying and studying algorithms based on the Darwinian evolution theory. Driven by selection (survival of the fittest, mating of the fittest) and randomised reproduction (mutation, recombination), an evolutionary process is being emulated and solutions for a given problem are being "bred". During this course, various flavours within evolutionary computing are treated, including genetic algorithms, evolution strategies, evolutionary programming, genetic programming, differential evolution, particle swarm optimisation. Applications in optimisation, constraint handling, machine learning, and robotics are discussed. Specific subjects handled include: genetic structures (representations), selection techniques, sexual and asexual reproduction operators, (self-)adaptivity and methodological aspects, such as algorithm design & tuning, and performance assessment. Special attention is paid to the field of evolutionary robotics where the bodies (morphology, hardware) and the brains (controller, software) of robots are evolved simultaneously. Hands-on-experience with evolutionary algorithms is gained through a compulsory programming assignment.

Additional Information Teaching Methods

Oral lectures and compulsory Python programming assignment (in teams of 3 to 5). A limited number of highly motivated students can replace the programming assignment by a special research assignment under the personal supervision of the lecturer(s). These research projects aim at publications with the students as first authors of the paper.

Method of Assessment

Written exam and programming assignment (in teams of 3 to 5). To pass the course as a whole, you must pass both the exam and the programming assignment. For the final grade they will count with the same weight:

Grade = 0.5 x Exam_Grade + 0.5 x Assignment_Grade

Note that:

- The exam is multiple choice, taken on the campus in an electronic form. There is a resit in case you cannot attend or fail.
- The programming assignment is graded based on your code, the quality of the solution your algorithm finds, and a short report. **No resit is possible for the programming assignment.**
- If you pass one component (Exam or Assignment), but fail the other, then you can keep the grade of the passed component for next year or later. Then you can pass the course by getting a good grade (> 5.5) for the missing component and using the old grade in the averaging formula.

Literature

The course is based on the textbook ***A.E. Eiben and J.E. Smith, Introduction to Evolutionary Computing, Springer, 2015, 2nd edition, ISBN 978-3-662-44873-1***. The lectures and the book chapters are synchronised to a great extent. In addition, a number of carefully selected papers to provide extra information on specific topics are available via Canvas.

Additional Information Target Audience

MSc Artificial Intelligence
MSc Econometrics and Operations Research
MSc Finance
MSc Business Analytics
MSc Computer Science
MSc Computer Security
MSc Bioinformatics and Systems Biology

Additional Information

Attending the lectures in real life is not mandatory, but highly recommended. Active participation in the team work for the programming assignment is a must!

Explanation Canvas

The complete schedule of the course, organised by the chapters of the textbook, will be made available at the beginning of the course. Students are assumed to consult the course schedule to keep up with the progress, as Canvas rubrics are not used. Video recordings of the lectures made during the COVID period are still available on Canvas for self-study.

Recommended background knowledge

Python programming skills are necessary to do the practical assignment. If you cannot program, you cannot complete this course.

Fundamentals of Bioinformatics

Course Code	X_405052
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. H. Mouhib, dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. D. Molenaar
Teaching method(s)	Seminar, Computer lab, Written partial exam, Lecture

Course Objective

Are you interested in Bioinformatics? Do you want to find out how biology can make an exciting application domain? Or do you want to learn what more you could do with your data, and with less effort? Enter here to start!

Fundamentals of Bioinformatics (FoB) is the starting course of the Bioinformatics master. It aims to give you a broad overview of important topics relevant to the field, with a focus on current open problems. You will be made aware of these open problems during practical sessions that will cover active research in a major Bioinformatics

problem (mutation impact prediction). Based on your background, you will be assigned to separate classes where you will be working to fill gaps in your background knowledge in programming and/or biology.

After completing the course successfully:

- You can explain the fundamental principles of cell biology and the genetic information flow
- You can write short code to solve simple problems and complete skeleton scripts for biological applications
- You can read algebra and calculus equations
- You can become aware of gaps in your own background knowledge.
- You can address gaps in your background knowledge (programming, mathematics and molecular biology) and skills
- You are aware of the major issues when developing and benchmarking new methodology in bioinformatics research
- You can present and communicate a joined group projects carried out with students of diverse backgrounds (Interdisciplinarity) in a respectful way
- You can apply and understand fundamental methods in bioinformatics, e.g. sequence alignment, impact prediction, and to understand and use fundamental concepts in bioinformatics, e.g. homology, evolutionary conservation, and gene function

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

Topics:

- Evolution, genomics, sequence alignment, Blast/PSI-Blast, FAIR data, benchmarking, machine learning, genome assembly, mutation impact prediction, protein stucture analysis, substitution matrices, scientific writing

Project:

- Group project to benchmark different mutation impact prediction approaches. The groups will be composed to include different BSc backgrounds.

Background knowledge (1 out of 3, see below):

- Molecular cell biology
- Programming and computational thinking in Python
- Calculus and linear algebra

Additional Information Teaching Methods

- 8 lectures (two-hour lecture in the morning, two days per week) Exercises during/in between lectures
- 12 computer practicals for the group project - compulsory attendance (online or on campus) (two hour sessions following the morning lectures, two days per week), supervised.
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Conversion classes for background knowledge:

- Molecular cell biology (7 x double lectures)
- Programming and computational thinking in Python (10 computer practicals)

Depending on your background, you will be assigned 1 of the 2 conversion classes. Students that are already well versed in both topics (molecular biology and programming) may be assigned to the following class, which is organised as part of the Introduction to Systems Biology:

- Calculus and linear algebra (10 maths classes)

Method of Assessment

- Project work [~40%] :
- Project report and/or presentation
- Digital exam to assess [~60%]:
- Lecture topics
- Concept learned during the project
- Concepts from conversion classes

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Further assessment and grading details will be posted on Canvas (resits and compensation rules)

Entry Requirements

Bachelor in any science discipline (including medicine) or strong programming background, and an interest in biological problems.

Literature

- course material (slides, scientific papers) on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics Garland Science 2008 ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will help you to set this up.

Recommended background knowledge

Programming in Python, Molecular Biology, some basics of calculus and linear algebra

Image Processing and Quantitative Data Analysis

Course Code	XM_0085
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Postma
Examiner	dr. M. Postma
Teaching Staff	dr. M. Postma
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

Knowledge and understanding:

- About the basic principles and modalities of several important bioimaging techniques and applications
- About different types of imaging data sets
- About different types of image processing and data analysis techniques
- About different types of software languages and tools for analysing image datasets

Applying knowledge:

- To be able to handle imaging datasets using ImageJ, Python and Maltab/Octave
- To be able to use and write scripting of analysis pipelines in relevant programming languages
- To be able to apply image processing techniques
- To be able to extract and analyse quantitative information from images
- To be able to apply knowledge from literature or other available sources

Making judgements:

- To select and validate the relevant image processing methods and tools in a correct way
- Based on a formulated bioimage-related research question, develop an approach, generate results, and interpret these results within the context of the research question
- Search, select and test methods found in the literature or other available sources

Communication:

- To write a report and/or to give a presentation about a research question, the analysis and results in a group of students.

Learning skills:

- To be able to understand and implement new methods from literature
- To create new approaches for specific applications and or research questions

Course Content

In biomedical and biological research many different (functional) imaging techniques exist, including fluorescence and optical microscopy (bright field, phase contrast, etc.), scanning electron microscopy (SEM), X-ray, ultrasound (sonography), magnetic resonance (MRI), computerised tomography (CT), positron emission tomography (PET). With these techniques we can visualise and study biological systems at the molecular, cellular, multicellular and tissue level in space and time. The datasets produced by these techniques are digital images and can be further used to extract quantitative information. The analysis of these 2D/3D or even higher dimensional datasets requires the use of image processing techniques.

In this course we will cover topics about quantitative bioimaging techniques and image analysis techniques including basic operations, region of interests, feature extraction, masks and the use of filters.

Furthermore we will cover more advanced processing techniques including image segmentation, spectral unmixing and color deconvolution, image registration, localisation and tracking, co-localisation. We will discuss and apply these techniques also within the context of widely used biomedical and biological applications, e.g. immunohistology, FISH, super resolution, FRAP, FRET, fluorescent labelling, cell tracking, enzymatic activity etc.

As a subsequent step quantified spatio-temporal information, for example about intensities, locations, timings and other parameters will be further statistically analysed, e.g. auto- and cross-correlation, point statistics, fitting, comparison and clustering.

Additional Information Teaching Methods

- About 8 lectures (2-3 two-hour lectures per week), exercises during/in between lectures and practicals
- About 8 computer practicals (2-3 afternoon sessions per week), feedback (theoretical and practical) will be given during and after the computer practical sessions
- Self-study
- Final project week presentation and critical discussion of state-of-the-art

Method of Assessment

Weights of the different examination components:

- 40% Group Project: Pipeline & Code(Github)
- 40% Group Project: Report (drylab journal)
- 20% Group Project: Presentation

Further assessment and grading details will be posted on Canvas (resits and compensation rules)

Entry Requirements

Bachelor in any science discipline with sufficient programming background

Literature

- Literature provided at the start or during the course
- Lecture slides provided at the start or during the course
- Practical training material provided at the start or during the course
- Software provided at the start or during the course

Additional Information Target Audience

M Bioinformatics and Systems Biology

Additional Information

Students are required to arrange and bring a laptop

Recommended background knowledge

Programming experience e.g. Python, ImageJ, Matlab

Introduction to Systems Biology

Course Code	X_428565
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G. Stolf Jeuken
Examiner	dr. D. Molenaar
Teaching Staff	dr. M.T.M. Wortel, dr. D. Molenaar, prof. dr. F.J. Bruggeman, dr. G. Stolf Jeuken
Teaching method(s)	Seminar, Study Group, Written partial exam, Lecture

Course Objective

Introduction to Systems Biology is the starting course of the Bioinformatics and Systems Biology master (together with Fundamentals of Bioinformatics). To emphasise the importance of systems thinking biology, this course introduces the principles of systems biology, with a primary focus on the analysis of metabolic networks using stoichiometric modelling.

The course explores how genomic information can be translated into computational models to understand and predict cellular behaviour, linking genotype and phenotype. The course will see the application of these models in biotechnology and medicine, utilizing mathematical concepts (particularly linear algebra) and computational tools.

After completing the course successfully [Bloom taxonomy]:

- You understand the challenge of linking genomic information (genotype) to cellular behaviour (phenotype). [understand]
- You can write short code to solve simple problems and complete skeleton scripts for computational modelling of biological processes [apply]
- You can solve mathematical problems using algebra. [apply]
- You are able to represent biological information, specifically metabolic networks, in a computable format using mathematical concepts. [evaluate]
- You are able to use computational tools and algorithms to analyse metabolic networks. [apply]
- You appreciate the importance of experimental data and design in systems biology. [analyse]

This will enable you to understand and perform computational analysis related to:

- Systems Biology Research: Using metabolic network models to understand cellular growth under different conditions.
- Biotechnological Applications: Identifying genes to modify in microbial genomes for the production of valuable products like biofuels.
- Medical Applications: Identifying potential drug targets within the metabolism of pathogens or cancer cells for therapeutic intervention.

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

The course will cover the following key topics:

- Genome Annotation and Network Reconstruction: Decoding genomic information to build metabolic networks.
- Stoichiometric Principles: Understanding chemical element and charge conservation in biochemical reactions, stoichiometric coefficients, and balancing reactions.

- Reaction Networks and Stoichiometric Matrix: Representing networks mathematically using the stoichiometric matrix and formulating mass balance equations.
- Steady-State Analysis: Concepts of steady-state, null space of the stoichiometric matrix, steady-state flux distributions, and the rank theorem.
- Linear Programming and Flux Balance Analysis (FBA): Introduction to FBA as a core computational method for analysing metabolism.

Conversion classes:

Depending on their needs, students can choose to attend conversion classes where they will be working to fill gaps in their knowledge in mathematics (linear algebra and differential equations), programming or biology.

Additional Information Teaching Methods

- Lectures
- Workgroups
- Conversion classes in biology or mathematics or programming

Method of Assessment

A written exam. To pass the exam, the grade must be 5.5 or higher. Retakes of the exam take place once a year. The last obtained pass grade will count as the final grade for the course.

Entry Requirements

Bachelor in any science discipline (including medicine), with an interest in applying mathematical techniques in biology. When in doubt please contact the coordinator.

Literature

A course syllabus will be provided

Additional Information Target Audience

Students with a bachelor degree in Biology, Bioinformatics, Mathematics, Physics or similar background

Recommended background knowledge

Mathematics at high-school level

Knowledge Representation

Course Code	XM_0059
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.S.H. Koopmann
Examiner	dr. P.S.H. Koopmann
Teaching Staff	dr. P.S.H. Koopmann
Teaching method(s)	Practical, Lecture

Course Objective

Knowledge and understanding: at the end of the course the students should be acquainted with the broad principles of knowledge representation, such as the separation of representation and reasoning, the declarative nature of representations, the universal (domain independent) nature of inference mechanisms.

Apply knowledge and understanding: students will have practical experience with different representation formalisms, and will be able to implement a reasoning tool for at least one of these formalisms. This will allow them to better understand the role of knowledge representation in the broader context of AI.

Making judgement: students will be able to set up empirical experiments in order to evaluate the pros and cons of Knowledge Representation formalisms in specific application areas.

Communication skills: students will be able to write a scientific report about an original research question in a small group of students.

Learning skills: students will be trained in acquiring knowledge about a set of complex formal systems, learn how to come up with a research question and scientific hypotheses, and perform the necessary (empirical) research to prove or disprove those hypotheses.

Course Content

We discuss important formalisms for Knowledge Representation and symbolic AI, such as classical logics, description logics and one additional relevant formalism dealing with non-montonic and/or quantitative reasoning.

Additional Information Teaching Methods

Lecture: 2 sessions of 2 hours each week, on demand a Q&A session every 2-3 weeks, Working Groups to work on exercises every 2-3 weeks, a combination of exercises and practical assignments. The practical assignment will involve implementation, experimental evaluation and writing a report.

Method of Assessment

- assignments in groups (total 50%),
- peer reviewing assignment reports (P/F)
- written exam (50%).

A minimum grade of 5.5 for each component is required. The peer reviewing assignment report is marked pass/fail only, but must be completed to finish the course (even for resit students).

There will be a resit of the written exam, there is no resit opportunity for the practical assignments.

Literature

The main source of information for the lecture will be the lecture and the slides - no additional literature research is mandatory.

Additional Information Target Audience

Master Artificial Intelligence
Master Bioinformatics and Systems Biology

Recommended background knowledge

1. Basic knowledge of logic (propositional logic and first-order logic)
2. A good working knowledge of programming in Python is required

Machine Learning for the Quantified Self

Course Code	XM_40012
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M. Hoogendoorn
Examiner	prof. dr. M. Hoogendoorn
Teaching Staff	V.G.M. Francois Lavet, prof. dr. M. Hoogendoorn, B. Borsboom
Teaching method(s)	Practical, Lecture

Course Objective

The main aims of this course are to make the student familiar with specific machine learning techniques for quantified self and sensory data. Both the theory as well as the practical side of matters will be addressed. More specifically, the following aims are distinguished:

1. Understanding the domain of the quantified self and sensory data
2. Knowing and being able to apply techniques for outlier detection on sensory data.

3. Knowing and being able to apply techniques for feature engineering on sensory data.
4. Knowing and being able to apply clustering techniques related to sensory data.
5. Understanding the theory of supervised learning and its implications.
6. Learning how to apply non-temporal machine learning techniques to sensory data.
7. Knowing and being able to apply temporal machine learning techniques to sensory data, including deep learning techniques.
8. Knowing and being able to apply reinforcement learning in the application of the quantified self.

The course if focused on the Dublin descriptors Knowledge and understanding (learning new relevant algorithm in this domain); Applying Knowledge and Understanding (since the aim is also to apply the algorithms in practice); Making judgements (which technique is appropriate, how to best apply the techniques for a specific case); Communication skills (how to report on your approach, choices and your results), and Learning skills (able to find new relevant techniques, assess their suitability, etc.).

Course Content

The quantified-self refers to large-scale data collection of a user's behavior and context via a range of sensory devices, including smart phones, smart watches, ambient sensors, etc. These measurements contain a wealth of information that can be extracted by means of machine learning techniques, for instance for the purpose of predictive modeling. In addition, machine learning techniques can be a driver for adaptive systems to support users in a personalized way based on the aforementioned measurements. The type of data does however require specialized machine learning techniques to fully exploit the information contained in the data. Examples of challenges include the temporal nature of the data, the variety in the type of data, the different granularity of various sensors, noise, etcetera.

In this course specific techniques to handle quantified self (or broader sensory data) will be treated. More in specific, it will address:

- Feature engineering (how do we come from raw data to usable features):
  * Removing noise from data
  * Handling missing data
  * Identifying (temporal) features
- Learning of user patterns:
  * Temporal machine learning approaches such as recurrent neural networks, time series analysis
  * Clustering approaches with dedicated distance metrics (including dynamic time warping)
- Adaptive feedback and support
  * Reinforcement learning
- Integration of the various components.

In addition, a number of real-life applications will be discussed. Next to lectures, there will be an extensive practical part, where students will learn to work with various algorithms work on a project they propose themselves.

Additional Information Teaching Methods

The course will be taught in four weeks. Both regular lectures (l, first two weeks) and practical sessions (p) will be organized. During the first two weeks the emphasis will be on the lectures and getting started with the practical assignment that ties closely to the content of the lectures. The third week will be working on the practical assignment fulltime. In the fourth week a mandatory final lecture will take place in which a selection of groups will present their project. The course ends with an exam.

Method of Assessment

Written exam (50%) and practical assignment (50%). For both parts, the grade needs to be sufficient to obtain a final grade. The practical assignment is done in groups of three. For the practical assignment, two intermediate versions of the report should be submitted for feedback that are given a pass/fail. Only in case of a pass for the two intermediate versions will a final grade be given for the practical assignment.

The written exam will be multiple choice and have a resit during the Summer period. On top, in case your assignment grade average is insufficient you are allowed to redo the assignments by improving it based on the feedback you received. The maximum grade for this resubmitted assignment is set to a 6.

Literature

Hoogendoorn, M. and Funk, B. Machine Learning for the Quantified Self - On the Art of Learning from Sensory Data, ISBN 978-3-319-66308-1, Springer Verlag 2018. The book can be downloaded free of charge through the UBVU, just follow the link to the book: http://dx.doi.org/10.1007/978-3-319-66308-1

Papers on state-of-the-art deep learning techniques.

Additional Information Target Audience

Master Artificial Intelligence, Master Business Analytics, Master Computer Science

Additional Information

Lecturer: Prof.dr. M. Hoogendoorn

Custom Course Registration

Registration for medical informatics (UvA) students: Please check the following document https://www.dropbox.com/s/a2kd7xacuxt5y8e/AI%20for%20Health%202022-2023.
and then register for secondary courses via https://vu.nl/en/student/information-about-registration-and-enrolment/in

Recommended background knowledge

Programming experience. Knowledge of basic machine learning algorithms.

Major Bioinformatics Research Project BSB

Course Code	XM_0103
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. M.L. van Daalen
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. M. Postma, dr. J.M. Urtasun Elizari, prof. dr. F.J. Bruggeman, dr. E. Tutucci, dr. ir. K.A. Feenstra, dr. D. Molenaar, dr. H. Mouhib, dr. E. Bosdriesz, dr. J.H. van Heerden, dr. A.R. Haydarlou
Teaching method(s)	

Course Objective

During the major research project, you will show that you can independently perform research, as part of fulfilling all exit qualifications of the Master Bioinformatics and Systems Biology.

You will have the opportunity to independently explore a complex scientific question, applying and expanding your skills in computational analysis, biological interpretation, and research design. You will engage in all stages of the research process—from formulating a research question and developing a strategy to analyzing data and communicating results. It is not only a demonstration of scientific ability but also a chance for you to grow as an independent researcher, capable of contributing meaningfully to the life sciences. This prepares you for a career as a researcher in industry as well as academia.

You can do your Major Research Project in many different types of locations; any research lab, e.g. in a company, university, institute, or hospital is possible. You can also do your research project abroad.

After successfully completing the research project,

- You can demonstrate deep knowledge and understanding of the context and background relevant to their research project. [knowledge and understanding]
- You can translate a scientific problem in a specific field into a well-defined research question using a suitable approach to investigate it. [applying knowledge and understanding]
- You can design and implement a coherent workflow or strategy involving the use of computational methods and/or quantitative experiments. [applying knowledge and understanding]
- You can integrate different types of knowledge (biological, mathematical, computational) to perform research in Bioinformatics [applying knowledge and understanding]
- You can manage complex data, involving retrieval, preprocessing, analysis and visualization, using tools appropriate to the problem. [applying knowledge and understanding]
- You can critically evaluate the strengths and weaknesses of their own research in the context of current scientific literature. [making judgements]
- You can document their work in a reproducible and transparent way (i.e., FAIR principles) using suitable tools and standards. [learning skills, communication skills]
- You can function effectively within a research environment, including planning and executing research within

deadlines, independent work, interdisciplinary collaboration, and professional communication. [learning skills, communication skills]
- You effectively communicate the results from their research project, both written in the research thesis, and orally in the presentation and defense [learning skills, communication skills]

Course Content

- You need to gain between 42-60 EC from a Minor and Major Research Project for your Bioinformatics and Systems Biology MSc degree.
- The Major has a minimum duration of 30 ECTS (5 months).
- The Major has a maximum duration of 60 ECTS (10 months).
- Minor and Major Research Projects must be on a different topic.
- The Major Systems Biology Research Project needs to be within the field of Systems Biology, and contain one of the main topics taught in the track or constraint choice courses, as well as a quantitative component focused on interpretation at the system level.
- Research projects should have sufficient modelling, data analysis and systems biology components, as determined by your VU/UvA examiner

For detailed information on the research projects please see: https://canvas.vu.nl/courses/36860/pages/research-projects.

Additional Information Teaching Methods

This course consists of an individual research project.

Method of Assessment

Formal assessment:

- Project proposal [Feedback + Pass/Fail]
- GO/NO-GO assessment [Feedback + Pass/Fail]
- Professional development [Feedback + Grade 20%]
- Research competence [Feedback + Grade 40%] (assessed based on thesis, presentation and defence)
- Research thesis [Feedback + Grade 30%]
- Presentations [Feedback + Grade 10%]

For more details, please see the Research Project Assessment Plan and Grading Rubrics provided on Canvas.

Entry Requirements

You cannot start your Major Bioinformatics Research Project before you have passed at least 42 EC (7 courses), of which at least 30 EC (5 courses) out of the compulsory courses or Bioinformatics specialisation courses. See the TER for the specific rules.

Additional Information Target Audience

This course is only open to students in the Master Bioinformatics and Systems Biology.

Custom Course Registration

Please make sure you register via VU.nl before you start this course.
You will need to (re)register for the course in the academic year you are aiming to hand in your thesis.

Major Systems Biology Research Project BSB

Course Code	XM_0104
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. M.L. van Daalen
Examiner	dr. ir. K.A. Feenstra

Teaching Staff	dr. E. Tutucci, dr. A.R. Haydarlou, dr. H. Mouhib, prof. dr. F.J. Bruggeman, dr. D. Molenaar, dr. ir. K.A. Feenstra, dr. J.M. Urtasun Elizari, dr. J.H. van Heerden, dr. E. Bosdriesz, dr. M. Postma
Teaching method(s)	

Course Objective

During the major research project, you will show that you can independently perform research, as part of fulfilling all exit qualifications of the Master Bioinformatics and Systems Biology.

You will have the opportunity to independently explore a complex scientific question, applying and expanding your skills in computational analysis, biological interpretation, and research design. You will engage in all stages of the research process—from formulating a research question and developing a strategy to analyzing data and communicating results. It is not only a demonstration of scientific ability but also a chance for you to grow as an independent researcher, capable of contributing meaningfully to the life sciences. This prepares you for a career as a researcher in industry as well as academia.

You can do your Major Research Project in many different types of locations; any research lab, e.g. in a company, university, institute, or hospital is possible. You can also do your research project abroad.

After successfully completing the research project,

- You can demonstrate deep knowledge and understanding of the context and background relevant to their research project. [knowledge and understanding]
- You can translate a scientific problem in a specific field into a well-defined research question using a suitable approach to investigate it. [applying knowledge and understanding]
- You can design and implement a coherent workflow or strategy involving the use of computational methods and/or quantitative experiments. [applying knowledge and understanding]
- You can integrate different types of knowledge (biological, mathematical, computational) to perform research in Systems Biology [applying knowledge and understanding]
- You can manage complex data, involving retrieval, preprocessing, analysis and visualization, using tools appropriate to the problem. [applying knowledge and understanding]
- You can critically evaluate the strengths and weaknesses of their own research in the context of current scientific literature. [making judgements]
- You can document their work in a reproducible and transparent way (i.e., FAIR principles) using suitable tools and standards. [learning skills, communication skills]
- You can function effectively within a research environment, including planning and executing research within deadlines, independent work, interdisciplinary collaboration, and professional communication. [learning skills, communication skills]
- You effectively communicate the results from their research project, both written in the research thesis, and orally in the presentation and defense [learning skills, communication skills]

Course Content

- You need to gain between 42-60 EC from a Minor and Major Research Project for your Bioinformatics and Systems Biology MSc degree.
- The Major has a minimum duration of 30 ECTS (5 months).
- The Major has a maximum duration of 60 ECTS (10 months).
- Minor and Major Research Projects must be on a different topic.
- The Major Systems Biology Research Project needs to be within the field of Systems Biology, and contain one of the main topics taught in the track or constraint choice courses, as well as a quantitative component focused on interpretation at the system level.
- Research projects should have sufficient modelling, data analysis and systems biology components, as determined by your VU/UvA examiner

For detailed information on the research projects please see: https://canvas.vu.nl/courses/36860/pages/research-projects.

Additional Information Teaching Methods

This course consists of an individual research project.

Method of Assessment

Formal assessment:

- Project proposal [Feedback + Pass/Fail]
- GO/NO-GO assessment [Feedback + Pass/Fail]

- Professional development [Feedback + Grade 20%]
- Research competence [Feedback + Grade 40%] (assessed based on thesis, presentation and defence)
- Research thesis [Feedback + Grade 30%]
- Presentations [Feedback + Grade 10%]

For more details, please see the Research Project Assessment Plan and Grading Rubrics provided on Canvas.

Entry Requirements

You cannot start your Major Systems Biology Research Project before you have passed at least 42 EC (7 courses), of which at least 30 EC (5 courses) out of the compulsory courses or Systems Biology specialisation courses. See the TER for the specific rules.

Additional Information Target Audience

This course is only open to students in the Master Bioinformatics and Systems Biology.

Custom Course Registration

Please make sure you register via VU.nl before you start this course. You will need to (re)register for the course in the academic year you are aiming to hand in your thesis.

Minor Research Project M BSB

Course Code	XM_0072
Credits	18
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. M.L. van Daalen
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. A.R. Haydarlou, dr. J.M. Urtasun Elizari, dr. D. Molenaar, dr. ir. K.A. Feenstra, dr. E. Bosdriesz, dr. H. Mouhib, prof. dr. F.J. Bruggeman, dr. J.H. van Heerden, dr. E. Tutucci
Teaching method(s)	

Course Objective

During the minor research project, you will show that you can independently perform research, as part of fulfilling all exit qualifications of the Master Bioinformatics and Systems Biology.

You will have the opportunity to gain hands-on research experience in a topic of their interest, either within or beyond the core areas of Bioinformatics and Systems Biology. This can serve both as preparation for the Major Research Project and as a valuable standalone experience. During the research you will independently explore a complex scientific question, applying and expanding your skills in computational analysis, biological interpretation, and research design. You will engage in many stages of the research process, including formulating a research question and developing a strategy to analyzing data and communicating results. It is not only a demonstration of scientific ability but also a chance for you to grow as an independent researcher, capable of contributing meaningfully to the life sciences. This prepares you for a career as a researcher in industry as well as academia.

You can do your Minor Research Project in many different types of locations; any research lab, e.g. in a company, university, institute, or hospital is possible. You can also do your research project abroad.

After successfully completing the research project,

- You can demonstrate deep knowledge and understanding of the context and background relevant to their research project. [knowledge and understanding]
- You can translate a scientific problem in a specific field into a well-defined research question using a suitable approach to investigate it. [applying knowledge and understanding]
- You can design and implement a coherent workflow or strategy involving the use of computational methods and/or quantitative experiments. [applying knowledge and understanding]
- You can integrate different types of knowledge (biological, mathematical, computational) to perform research in Bioinformatics or Systems Biology or a related topic [applying knowledge and understanding]

- You can manage complex data, involving retrieval, preprocessing, analysis and visualization, using tools appropriate to the problem. [applying knowledge and understanding]
- You can critically evaluate the strengths and weaknesses of their own research in the context of current scientific literature. [making judgements]
- You can document their work in a reproducible and transparent way (i.e., FAIR principles) using suitable tools and standards. [learning skills, communication skills]
- You can function effectively within a research environment, including planning and executing research within deadlines, independent work, interdisciplinary collaboration, and professional communication. [learning skills, communication skills]
- You effectively communicate the results from their research project, both written in the research thesis, and orally in the presentation and defense [learning skills, communication skills]

Course Content

- You need to gain between 42-60 EC from a Minor and Major Research Project for your Bioinformatics and Systems Biology MSc degree.
- The Major has a minimum duration of 30 ECTS (5 months).
- The Major has a maximum duration of 60 ECTS (10 months).
- Minor and Major Research Projects must be on a different topic.
- The Major Systems Biology Research Project needs to be within the field of Systems Biology, and contain one of the main topics taught in the track or constraint choice courses, as well as a quantitative component focused on interpretation at the system level.
- Research projects should have sufficient modelling, data analysis and systems biology components, as determined by your VU/UvA examiner

For detailed information on the research projects please see: https://canvas.vu.nl/courses/36860/pages/research-projects.

Additional Information Teaching Methods

This course consists of an individual research project.

Method of Assessment

Formal assessment:

- Project proposal [Feedback + Pass/Fail]
- GO/NO-GO assessment [Feedback + Pass/Fail]
- Professional development [Feedback + Grade 20%]
- Research competence [Feedback + Grade 40%] (assessed based on thesis, presentation and defence)
- Research thesis [Feedback + Grade 30%]
- Presentations [Feedback + Grade 10%]

For more details, please see the Research Project Assessment Plan and Grading Rubrics provided on Canvas.

Entry Requirements

You cannot start your Minor Research Project before you have passed at least 18 EC (3 courses) out of the compulsory courses or specialisation courses. See the TER for the specific rules.

Additional Information Target Audience

This course is only open to students in the Master Bioinformatics and Systems Biology.

Custom Course Registration

Please make sure you register via VU.nl before you start this course.
You will need to (re)register for the course in the academic year you are aiming to hand in your thesis.

Protein Science

Course Code	AM_470145
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. D. Bald
Examiner	dr. D. Bald
Teaching Staff	dr. D. Bald, dr. H. Mouhib, dr. ir. Y.J.M. Bollen, dr. D.P. Geerke
Teaching method(s)	Study Group, Lecture

Course Objective

The student:

1. knows and understands principles of protein structure, function, dynamics, regulation, inhibition, interaction and engineering
2. can explain and predict protein function based on protein structure and understanding the molecular properties of amino acid residues.
3. knows and understands the principle of current methods for protein investigation (e.g. structure determination, activity assays, overproduction, purification, interaction)
4. can analyze the strong and weak points of Protein Science techniques and can correlate an open question with a suitable technique/approach.
5. can analyze experiments in Protein Science and design new experiments.

Course Content

We will start with a repetition of protein structure and function.

Subsequently, we will focus on methods in protein science and also on more specialized properties of proteins important in fundamental research, biomedicine or biotechnology. Finally we will deal with case studies on selected proteins.

Planned lecture topics include:

- Protein Structure, Protein Function, Protein Dynamics, Control of
- Protein Function, Antibiotic action and antibiotic resistance,
- Protein over-production and purification, Protein structure and interaction, Protein Engineering.

Planned case studies:

- New antibiotics, Protein hydrophobicity and folding,
- Molecular modeling/docking, Protein-protein Interaction modulation.

Additional Information Teaching Methods

Lectures and work discussions, accompanied by self study to prepare for the lectures and to discuss the material presented in lectures.

Method of Assessment

Written exam (100%)

Entry Requirements

Participation in prior Bachelor-level courses on proteins/biochemistry is expected.

Literature

No specific book required. You can use any Biochemistry textbook, e.g. "Voet and Voet", for repetition.

A more concise alternative is "Protein Structure and Function" by Petsko/Ringe (somewhat old, but nicely written).

You will receive reading/study material, mainly reviews and original articles on relevant topics.

Examples of scientific literature: Jin et al, Nature 2020, Goojani et al. Biochim. Biophys. Acta 2020, Ple et al. Nature Communications 2022.

Additional Information Target Audience

The main target audience are Masters students Biomolecular Sciences. Additionally, students from Masters programs Biomedical Sciences, Bioinformatics, Pharmaceutical Sciences and Medical Natural Sciences can participate.

Additional Information

This course follows the VU guidelines on the use of generative AI in education (https://vu.nl/en/student/examinations/generative-ai-your-use-our-expectations)

Recommended background knowledge

Participation in prior Bachelor-level courses on proteins/biochemistry is expected.

Quantitative Single-cell Biology

Course Code	XM_0084
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.J. Bruggeman
Examiner	prof. dr. F.J. Bruggeman
Teaching Staff	dr. E. Tutucci, prof. dr. F.J. Bruggeman
Teaching method(s)	Seminar, Computer lab, Practical, Lecture

Course Objective

Motivation - Modern eukaryotic cell biology, microbiology and molecular systems biology, and their applications in medicine, rely heavily on single-cell approaches. Live-cell imaging with fluorescence microscopy, flow cytometry and single-cell sequencing are becoming standard tools in labs. With it come theoretical challenges too. Methods from statistical biophysics, quantitative image analysis and machine learning are being used to predict, analyze raw data, and understand experimental outcomes. Thus, researchers need to be trained in the experimental and theoretical aspects of quantitative single-cell biology.

After successful completion of this course:

- You can give an overview of the most modern single-cell analysis theoretical and practical method
- You can explain the key facets of quantitative single-cell analysis
- You can explain the basic theory and simulation methods in addition to experimental single-cell analysis methods, which include single-molecule RNA imaging (smFISH), fluorescent-tagging of proteins, fluorescence microscopy and flow cytometry
- You can use and adapt custom-written Python code to analyze single cell data
- You can create new analysis approaches for specific research questions and can use and write scripting of analysis pipelines
- You can analyze experimental data (imaging analysis of fluorescence microscopy data, flow cytometry and growth measurements)
- You can investigate how single-cell microorganisms adapt to changing environments
- You can explain concepts including fitness, gene expression noise, adaptation of metabolism and population heterogeneity
- You can use computational methods to analyze the data (i.e. quantify single-cell RNA and protein expression changes in space and time), which will be compared to expectations deriving from theory and stochastic simulations
- You can report the result of their analysis in a research paper-like format to demonstrate your ability to convey complex biological question and approaches
- You are familiar with the basic concepts, successes, and challenges of the current field of quantitative single-cell biology
- You are able to analyze complex biological questions

Course Content

The main topics covered by the course are as follows:

- Key single-cell phenomena from cell birth to division, single-cell stochasticity;
- Relations between microbial fitness, dynamic environments, phenotypic heterogeneity and phenotypic diversification;
- Modern Single-cell techniques, e.g. microscopy (single molecule RNA imaging and fluorescent protein tagging) and flow cytometry;
- Data analysis: stochastic simulation, image analysis, cell-segmentation, cell-tracking and statistical methods

using a combination of Mathematica, Phython and ImageJ;
- Experimental methods: fluorescent protein labeling, single-molecule RNA imaging methods, real-time imaging of cell growth and fluorescent reporters; flow cytometry.

Additional Information Teaching Methods

Teaching Methods

- Lectures (Week 1: 14 hours; Week 2: 8 hours; Week 3: 6 hours)
- Work groups, including data analysis, report writing and Q&A sessions (Week 1: 12 hours; Week 2: 21 hours; Week 3: 21 hours; Week 4: 7.5 hours)
- Lab visit (1 afternoon)

Method of Assessment

Theoretical exam (70%) and project (30%).

Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Basic cell biology, basic statistics and basic computer skills.

Literature

Course syllabus and PowerPoints provided during the course.

Additional Information Target Audience

MSc Bioinformatics and Systems Biology
MSc Biomolecular sciences

Additional Information

Students are required to bring a laptop

Scientific Machine Learning

Course Code	XM_0138
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G. Stolf Jeuken
Examiner	dr. H. Mouhib
Teaching Staff	dr. P. Bloem, dr. H. Mouhib, dr. G. Stolf Jeuken
Teaching method(s)	Seminar, Lecture

Course Objective

Upon completion of this course, students will:

- be acquainted with the dominant concepts of machine learning methods, including some theoretical background. (Knowledge and understanding)
- acquire knowledge of established machine learning techniques such as linear models, neural networks, decision trees and ensemble methods (Knowledge and understanding)
- learn some statistical techniques to assess and validate machine learning results. (Apply knowledge and understanding, make judgments)
- have worked with a scientific data set related to bioinformatics to define and solve a relevant scientific ML and biological question (Apply knowledge and understanding, make judgement and communicate with fellow students and teachers)

Course Content

Machine learning is the discipline that studies how to build computer systems that learn from experience. It is a subfield of artificial intelligence that intersects with statistics, cognitive science, information theory, and probability theory.

Recently, machine learning has become increasingly important for the design of search engines, robots, and sensor systems, and for the processing of large scientific data sets. It has also become increasingly important to tackle bioinformatics questions related to diseases, molecular interactions, and molecular property predictions.

The course covers a wide variety of machine learning techniques, but puts particular emphasis on gradient descent optimization, backpropagation, neural networks and deep learning. Some discussion on the broader social impact of machine learning technology is included.

During the practical part, students will work together on real world data sets from cancer research, protein-protein interactions, protein aggregation and solubility, or small bio-active molecules. Instead of focusing solely on improving model prediction, the ML methods are leveraged to investigated the provided data sets and gain a better understanding of the underlying biological problems.

Additional Information Teaching Methods

The course evaluation consists of two parts: an examination and a practical assignment (50%). The examination consists of a standard exam (50%) and four online quizzes (pass/fail - required for the exam).

The examination is supported by pre-recorded videos, interactive lecture/QA sessions (two per week) and optional homework assignments discussed in working groups (one per week). The practical assignment is supported by small exercises to help with the relevant technologies, and informal presentations at project groups (one per week).

There is no mandatory attendance for any lectures, however, the weekly project group meetings are mandatory for all group members to actively participate in the discussion of the other presented group projects. Every student is allowed to skip one group project session, as long as at least one member of the group is present during the project group.

A large amount of the material is freely available at http://mlvu.github.io

The course is taught in English.

Method of Assessment

The examination and practical assignment both comprise 50% of the final grade. To pass the course, the examination grade should be at least 5.5, the practical assignment grade should be at least 5.5 and the average should be at least 5.5.

The examination is made individually, and the practical assignment is made in groups of 5. There is a resit for the exam, no resit is possible for the practical assignment.

Literature

There is no textbook. Some reading material will be provided digitally.

Additional Information Target Audience

The course project uses a selection of preprocessed data sets tailored for the Master Bioinformatics and Systems Biology.

Please note that students who completed Machine Learning (X_400154) as part of their Bachelor programme, or who completed this course during the Master before 2024-2025, are not allowed to take Scientific Machine Learning (XM_0138) as part of their Master programme.

Custom Course Registration

Working group registration will be done through Canvas in the first week of the course. Only a registration for the main course itself is required to take part in all parts of the course.

Recommended background knowledge

We require that students have some prior experience with linear algebra, calculus (limited to differentiation), and probability theory or statistics.
A basic understanding will suffice and we will take some time to go over the basics again. Feel free to register if you have no experience with any of these, but expect to put in a little extra effort in the first weeks (using the materials provided).

Programming experience, preferably in Python, is highly recommended.

For the project, an interest in bioinformatics-related topics such as cancer research, protein-protein interactions, protein aggregation and solubility, or small bioactive molecules is highly recommended.

Statistics with R

Course Code	X_418156
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Molenaar
Examiner	dr. D. Molenaar
Teaching Staff	dr. G. Stolf Jeuken, dr. D. Molenaar
Teaching method(s)	Study Group, Computer lab, Lecture

Course Objective

Biologists and biomedical scientists often need to handle and analyse large sets of biological data. A popular tool for such analyses is the open-source statistical programming environment R. R is used intensively in the community of experimental biologists and biomedical scientists, and most newly published data analysis techniques are first available as R-packages.

This course focuses on obtaining the practical skills to perform data handling and analysis tasks from small to large data sets, and to graphically display and interpret the results. Statistical analyses will center on multivariate analysis and causal reasoning.

After completing the course successfully:

- You can use the R-language for reading, wrangling and analysis of data [applying knowledge and insight]
- You can create reports with integrated code, graphs, tables and text [applying knowledge and insight, communication]
- You know techniques of data stewardship and reproducible research [knowledge and insight]
- You can apply basic concepts of probability theory to problems in data analysis and bioinformatics [applying knowledge and insight]
- You can apply linear modeling and ANOVA or model selection criteria like BIC to investigate statistical relations between variables [applying knowledge and insight]
- You can apply logistic regression, naïve Bayes and decision trees to build classifiers [applying knowledge and insight]
- You can interpret and draw causal graphs and infer statistical relations between variables from causal relations [judgement, applying knowledge and insight]
- You can critically evaluate the role of data analysis in the experimental cycle [judgement]
- You can apply linear and other modeling techniques to perform causal inference [applying knowledge and insight]

Course Content

The first part of the course will focus on training the technical aspects of the R-language and of tools for reproducible working.

- Programming effectively in the R language, knowing the basic data structures and their relations, constructing graphs using *ggplot* and apply data manipulation techniques using *tidyverse* functions.
- Document generation with integrated code, cross references and literature citations, and techniques for reproducible calculation.

In the second part we will apply R in probability, statistical modelling and machine learning using examples from various fields of biology, ranging from cell biology to ecology.

- Probability theory: applying the basic concepts of probability theory: sample and event spaces, axioms and theorems of probability, random variables, probability distributions, conditional and joint probability, dependence, Bayes rule, expected values.
- Statistical modeling: create and analyse statistical models, and knowing its basic concepts: linear modeling, techniques of model selection and model comparison like ANOVA, cross validation, information criteria.
- Causal inference: knowing how to interpret and construct causal diagrams for a data generating process, and

knowing how to perform causal inference from causal diagrams and data.

Additional Information Teaching Methods

On campus lectures, work groups, none being mandatory.

Method of Assessment

- Assessment will take place by means of two individual assignments as well as a written exam. Note that these assignments take place on fixed dates and each take approximately a working day to complete. The dates will be announced well before the start of the course.
- To pass the course, the average of the weighted scores on the assignments must be 5.5 or higher and the score on the exam must be 5.5 or higher. The final grade is determined by the scores on the assignments (70%) and the exam (30%).
- Mode of re-examination: one assignment on a similar subject and with the same weight as the last assignment can be submitted once. Also, a re-take of the exam takes place once a year. In all cases the last obtained "pass" score per assessment component (assignments or exam) will be used to calculate the final grade.

Entry Requirements

Bachelor in any science discipline (including medicine). Knowledge of statistical hypothesis testing is required. Some experience with programming is highly recommended. When in doubt please contact the coordinator.

Literature

A course syllabus with theory and exercises is available online.

Additional Information Target Audience

- Master Bioinformatics and Systems Biology
- Master Biomolecular Sciences

Recommended background knowledge

Knowledge of basic statistics is essential. Some experience with a programming language is highly desirable.

Structural Bioinformatics

Course Code	X_405019
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	R. Kumar, dr. ir. K.A. Feenstra, dr. J.M. Urtasun Elizari, dr. H. Mouhib
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

Why Structural Bioinformatics?

The function of a protein is determined by its three-dimensional structure, and therefore structural information is crucial for understanding the working of proteins. However, experiments, prediction and simulation of protein structures remain difficult. This course will provide you with an overview of existing computational techniques, to validate, simulate, predict and analyse protein structures. More importantly, it will provide practical knowledge about how and when to use such techniques.

After completing the course successfully,

- You understand the basics of statistical thermodynamics and the relation to protein structures, evolution and function [Knowledge and understanding]
- You are able to evaluate protein structures with knowledge of their experimental source, bias and validation

techniques [Applying knowledge and understanding]
- You are able to compare different protein structures, and evaluate similarity [Applying knowledge and understanding]
- You know how and when to use structure prediction methods [Making judgements; Applying knowledge and understanding]
- You can perform calculations on protein structures to gain new biological insights [Applying knowledge and understanding]
- You are able to compare different simulation techniques for biological macro-molecules, and be able to analyse the simulated data computationally [Applying knowledge and understanding]
- You are able to read and understand scientific papers in the field of Structural Bioinformatics [Communication; Lifelong learning skills]
- You are able to independently search for answers to open problems and ask a question about something that is unclear [Communication, Lifelong learning skills]

Course Content

Theory:

- Protein and DNA structure sources
- Experimental methods
- Structure validation
- Protein fold prediction (from homology modelling to ab initio prediction)
- Structural classification and structural alignment
- Protein folding and energetics
- Molecular Dynamics & Monte Carlo simulation
- Function from structure

Practical:

- Obtaining geometric features from PDB files
- Homology modelling with Modeller
- Protein interaction as a 'computational experiment' (simulation)

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- Exercises during/in between lectures
- 12 computer practicals (2 two-hour sessions per week)
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Method of Assessment

The final grade for this course will consist of 50% practical work and 50% theoretical assessment.

Practical Assignments: (50%)
(assessment of practical assignments will be adjusted based on ongoing developments in generative language models)

Theoretical: (50%)

- Oral, written or digital exam (depending on number of course students) to assess lecture topics.
- As part of the exam research papers on Structural Bioinformatics topics need to be analysed in detail.
- You will be prepared for your exam through exercises and paper discussions during the lectures, as well as through the practical assignments.

Assessment and grading details will be posted on Canvas.

Entry Requirements

Bachelor in any science discipline (including medicine), with an interest in molecular biology and applying algorithmic approaches to protein structures.
Experience with programming (preferably python).
Basic understanding of calculus and of organic chemistry.
When in doubt please contact the coordinator.

Literature

- course material on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum. Understanding Bioinformatics. Garland Science 2008 ISBN-10:

0-8153-4024-9
- book "Introduction to Structural Bioinformatics": The intro is published on ArXiv [https://arxiv.org/abs/1801.09442], and the chapters are available from the table of contents. The book will also be posted on canvas.

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will provide some help in setting this up at the start of the course.
Lectures may be recorded; these recordings will only be used within the course, including possibly in subsequent years as background information.

Recommended background knowledge

Some experience with the linux command line is preferred.

Systems Biology in Practice

Course Code	XM_0079
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. F. Branco dos Santos
Examiner	dr. F. Branco dos Santos
Teaching Staff	dr. F. Branco dos Santos
Teaching method(s)	Seminar, Computer lab, Practical, Lecture

Course Objective

The aim of the course is to get acquainted with the interdisciplinary approach of experimental microbial physiology and -omics analysis.

Upon completion of this course, students will:

- learn how information obtained by experiments at the level of cellular behaviour, genetic profile and enzymatic make-up can be combined in order to get insight in the mechanisms underlying regulation and adaptation of microbial organisms;
- become familiar with the basic techniques and principles of microbial physiology and transcriptome analysis;
- become familiar with the basic techniques and principles and data handling and analysis of masspectrometry data;
- learn how to perform proteomic and transcriptomic data analysis;
- learn how to interpret and make conclusions from the data analysis;
- learn and practice how to discuss and present own findings in written and oral settings through assignments;
- foster team work and collaboration skills with group assignments.

Course Content

Throughout the course, the same experimental material will be used. Microorganisms will be cultured under a set of different conditions in the Molecular Microbial Physiology Group and their physiology will be quantified in terms of metabolic fluxes. Transcriptome analysis of the cells will show how and to what extent the environment affects the metabolic network at the genetic level. Finally the same cells will be analysed at the level of their protein composition by a proteomic analysis and by enzyme activity assays in vitro. Lectures on more general subjects of Microbial Systems Biology will be organised to give the students a flavor of practice and application.

The main topics covered during the course can be summarised as follows:

- Analyzing Bacterial Growth and Metabolic Flux - The student can understand and apply concepts of bacterial growth, flux analysis, and biochemical assays in a laboratory setting;
- Exploring Genetic Tools and Gene Expression - The student can explain and utilize genetic tools for gene expression studies, including transcriptomics techniques and their applications;
- Understanding and Applying Mass Spectrometry in Proteomics - The student can perform and analyze proteomics data using mass spectrometry, understanding the principles and practical applications of protein identification;
- Conducting Data and Systems Biology Analysis - The student can analyze protein data and understand systems biology approaches to study microorganisms, integrating various data types for comprehensive analysis;
- Performing Computational Analysis of Molecular Profiling - The student can conduct computational analysis on molecular profiling techniques including whole-genome sequencing (WGS), whole-exome sequencing (WES), proteomics, RNA sequencing, and array comparative genomic hybridization (arrayCGH);
- Applying Computational Methods to Data Analysis - The student can apply machine learning, normalization, regularization, feature selection, classification, read mapping, and clustering methods to analyze biological data;
- Developing and Assessing Models in Computational Biology - The student can perform cross-validation, critically assess model performance, and understand the clinical relevance of these analyses for diagnosis, treatment decisions, or biomarker discovery;
- Engaging in Professional Development and Collaborative Work - The student can work effectively in a group, write comprehensive reports, peer-review others' work, and deliver presentations, enhancing their professional and collaborative skills.

Additional Information Teaching Methods

- Lectures
- Seminars
- Practicals

Method of Assessment

Written thesis, written lecture abstracts and oral presentations.
Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Literature

Literature:
Lecture contents (available on Canvas).